ZETA ALONG A SEMICIRCLE

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Abstract. A question on Mathematics Stack Exchange asks whether $|\zeta(z)|$ increases strictly along the semicircle $z = 1 - 5e^{iu}$, $0 \le u \le \pi$, which runs from the trivial zero -4 through the critical strip to 6. We prove that it does, with a computer-assisted argument in ball arithmetic, and that the same holds for the semicircle of radius 3 through -2. For the semicircles of radius $7, 9, \dots, 41$ through the other trivial zeros it fails: $|\zeta|$ decreases on a certified interval. A heuristic from the functional equation explains the threshold: once the arc reaches height 2π while moving to the right in the half-plane $\operatorname{Re} z < 0$, $|\zeta|$ falls. The exact reductions behind the computation are checked in Lean.

1. The question

Start at the trivial zero $z = -4$ of the Riemann zeta function and walk to $z = 6$ along the lower semicircle centred at 1, $z(u) = 1 - 5e^{iu}$ for $0 \le u \le \pi$. A Mathematics Stack Exchange user asked whether $|\zeta(z(u))|$ increases strictly along the way, and whether a computer-assisted proof would count [4]. The walk starts at $|\zeta| = 0$ and ends at $\zeta(6) = \pi^6/945 \approx 1.017$, but in between it crosses the critical strip, where $|\zeta|$ can do anything. Before reading on, it is worth guessing what happens on the larger semicircles through -6, -8, ...: Figure 1 gives the answer away.

Theorem 1. *For $r = 3$ and $r = 5$, the function $u \mapsto |\zeta(1 - re^{iu})|$ is strictly increasing on $[0, \pi]$.*

Theorem 2. *For every odd r with $7 \le r \le 41$, the function $u \mapsto |\zeta(1 - re^{iu})|$ is not monotone on $[0, \pi]$.*

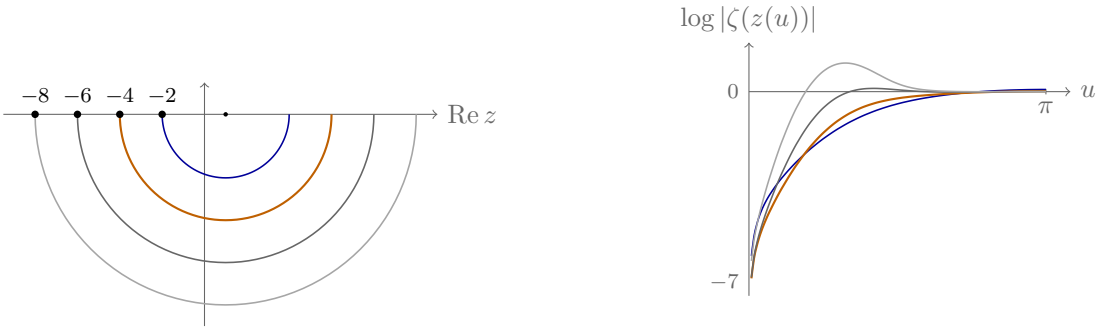


Figure 1. Left: the semicircles $z = 1 - re^{iu}$ for $r = 3, 5, 7, 9$, from the trivial zero $1 - r$ to $1 + r$. Right: $\log|\zeta|$ along them. For $r = 3$ (blue) and $r = 5$ (copper, the question) it rises all the way; for $r = 7$ (dark grey) and $r = 9$ (light grey) it overshoots and falls before the end.

Both theorems are computer assisted. On the second question in [4]: such a proof is a proof, provided every number is an enclosure with a proven error bound rather than a floating-point estimate. All computations below use Arb [1] ball arithmetic through python-flint [2], and every quantity is enclosed over a whole interval of u at once.

2. The reduction

Write $F(u) = \log|\zeta(z(u))|$ and $z(u) = 1 - re^{iu}$, so $z'(u) = -rie^{iu}$ and $|z'| = r$. Away from zeros of ζ,

$$R(u) := F'(u) = \mathrm{Re}\Big(z'(u)\,\frac{\zeta'}{\zeta}\big(z(u)\big)\Big),$$

because $\log|\zeta|$ is the real part of a branch of $\log\zeta$. So it suffices to show $R > 0$ on $(0, \pi)$. Two points need care, and each has a short exact argument.

The start. At $u = 0$ the path sits on the zero $z_0 = 1 - r$, so $R \to +\infty$ there and cannot be enclosed directly. Write $\zeta(z) = (z - z_0)h(z)$ with

$$h(z) = \int_0^1 \zeta'\big(z_0 + s(z - z_0)\big)\,ds, \qquad h'(z) = \int_0^1 s\,\zeta''\big(z_0 + s(z - z_0)\big)\,ds.$$

Then $\zeta'/\zeta = 1/(z - z_0) + h'/h$, and since $z - z_0 = r(1 - e^{iu})$ a short computation gives $\mathrm{Re}\big(z'/(z - z_0)\big) = \frac{1}{2}\cot\frac{u}{2}$. Hence

$$R(u) \;\geq\; \tfrac{1}{2}\cot\tfrac{u}{2} - r\Big|\frac{h'}{h}\Big|. \tag{1}$$

If B is a convex box containing z_0 and the arc for $0 \leq u \leq \delta$, then $h(z)$ is an average of values of ζ' on B, and $h'(z)$ is half an average of values of ζ'' on B. So h and h' lie in the boxes enclosing $\zeta'(B)$ and $\frac{1}{2}\zeta''(B)$, and (1) with $\frac{1}{2}\cot\frac{u}{2} \geq \frac{1}{2}\cot\frac{\delta}{2}$ proves $R > 0$ on $(0, \delta]$ once these enclosures are tight enough. In fact $h'/h \approx \zeta''(z_0)/(2\zeta'(z_0))$ is small: for $r = 5$ the enclosures give $5|h'/h| \leq 3.34$, against $\frac{1}{2}\cot 0.025 \approx 19.99$ for $\delta = 0.05$.

The end. At $u = \pi$, $z = 1 + r$ and $z' = ri$, while $\zeta(1 + r)$ and $\zeta'(1 + r)$ are real; so $R(\pi) = 0$ exactly. The arc meets the real axis at right angles, and by the symmetry $\zeta(\bar{z}) = \overline{\zeta(z)}$ the point $u = \pi$ is a turning point of F. So R cannot be bounded away from 0 near π. Instead we show $R' < 0$ on $[\pi - \varepsilon, \pi]$, where

$$R'(u) = \mathrm{Re}\Big(z''\frac{\zeta'}{\zeta} + z'^2\Big(\frac{\zeta''}{\zeta} - \Big(\frac{\zeta'}{\zeta}\Big)^2\Big)\Big), \qquad z'' = re^{iu};$$

then $R(u) = -\int_u^\pi R' > 0$ for $u < \pi$.

The middle. On $[\delta, \pi - \varepsilon]$ we enclose R on intervals of u, bisecting until every enclosure is positive.

3. The computation

With $\delta = 0.05$, $\varepsilon = 0.3$ and 64-bit balls, the three steps succeed for both radii:

r	intervals with $R > 0$ on $[0.05, \pi - 0.3]$	intervals with $R' < 0$ on $[\pi - 0.3, \pi]$
3	1695	3
5	3219	3

Each run takes about a second. This proves Theorem 1. The margin is thin near the top: for $r = 5$, on the last quarter of the arc $|\zeta|$ rises only from about 0.963 to 1.017, and $R(u)/\sin u$ tends to about 0.18 as $u \to \pi$. The left half of the arc, through the critical strip and the half-plane $\mathrm{Re}\, z < 0$, is where ball arithmetic does the real work.

For Theorem 2 it is enough to find, for each r, one interval of u on which the enclosure of R is negative. Scanning intervals of length $\pi/4000$ finds one for every odd r from 7 to 41; for $r = 7$ it is $[1946\pi/4000, 1947\pi/4000]$, near $u = \pi/2$.

Remark 3 (Why the threshold lies between 5 and 7). For $\mathrm{Re}\, z = \sigma < 0$ and large $|t| = |\mathrm{Im}\, z|$, the functional equation and Stirling's formula give $|\zeta(\sigma + it)| \approx (|t|/2\pi)^{1/2-\sigma}|\zeta(1 - \sigma - it)|$ [5], and the last factor is close to 1. So, at fixed height, $|\zeta|$ *decreases* as σ increases once $|t| > 2\pi$. Near $u = \pi/2$ the arc moves almost horizontally to the right at height about r while $\sigma = 1 - r\cos u$ is still negative for $u < \arccos(1/r)$, so for r well above $2\pi \approx 6.28$ the factor $(|t|/2\pi)^{-\sigma}$ wins and F falls. For $r = 3$ and $r = 5$ the arc never reaches height 2π. This is a heuristic, not a proof; turning it into a proof for all $r \geq 7$ needs explicit error terms.

4. Verification

The script `prove.py` proves the case $r = 5$ asked in [4], and `semicircles.py` proves both theorems; both use only ball arithmetic. Planted errors are caught: claiming monotonicity past $u = \pi$ (where it fails by symmetry), claiming monotonicity for $r = 7$, and looking for a decrease for $r = 3$ all fail to certify. A floating-point evaluation of $R(1)$ in mpmath agrees with the enclosure to 15 digits.

The exact reductions are formally verified in Lean 4 with Mathlib [3]: the identity $\operatorname{Re}\big(z'/(z - z_0)\big) = \frac{1}{2}\cot\frac{u}{2}$, the splitting $\big((z - z_0)h\big)'/\big((z - z_0)h\big) = 1/(z - z_0) + h'/h$, the bound (1) in the form used, $R(\pi) = 0$ for real values of ζ and ζ' at $1 + r$, and the step from $R' < 0$ and $R(\pi) = 0$ to $R > 0$. The enclosures of ζ, ζ' and ζ'', the averaging argument and the formula for R' are outside Lean. Code and Lean files are at [6].

Acknowledgments

The question is due to the Mathematics Stack Exchange user shown as user45639; the use of $\operatorname{Re}(z'\zeta'/\zeta)$ follows a comment of Greg Martin on the question. The computations and the Lean formalisation were carried out with Claude (Anthropic). The author is responsible for the content.

References

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[5] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.
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